

3.7 Ratio Test Worksheet

Determine whether each of the following series converge or diverge. You may use any theorem from 3.7 or earlier.

1. $\sum_{k=1}^{\infty} \frac{k^3}{3^k}.$

2. $\sum_{k=1}^{\infty} \frac{2^{2k+1}}{3^k}$

3. $\sum_{k=1}^{\infty} \frac{(2k)!}{4^k}$

4. Consider the series $\sum_{k=1}^{\infty} \frac{1}{2k-1}$.

i. What does the ratio test conclude about the series?

ii. Use any other method to determine whether or not the given series diverges.

$$5. \sum_{k=1}^{\infty} \frac{2 + (-1)^k}{5^k}$$

$$6. \sum_{k=1}^{\infty} \frac{k}{2^k}$$

$$7. \sum_{k=1}^{\infty} \frac{(-1)^k \cdot k}{k^2 + 1}$$

8. $\sum_{k=1}^{\infty} \frac{k^2 + 2}{k^4}$

9. Classify each of the following series as absolutely convergent, conditionally convergent, or divergent.

a) $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^2 + 3}$

b) $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} (n!)^2}{(2n)!}$